

Mechanics Bonus Problem

Budapest University of Technology and Economics

Problem Set 3 — Spring 2026

Exercise 1: Mission Imbuoyable

(20 pts)

There is a circular opening with a small rim at the bottom of a vessel. A wooden cube with side length $a = 10\text{ cm}$ and density $\rho_{\text{wood}} = 600\text{ kg m}^{-3}$ is placed on this rim. While pressing the cube downward, water is poured into the vessel until the upper face of the cube is covered by a water column of height $h = 5\text{ cm}$. After this, the cube is released, but it does not rise out of the water. The density of water is $\rho_{\text{water}} = 1000\text{ kg m}^{-3}$. Determine the radius of the opening at the bottom of the vessel.

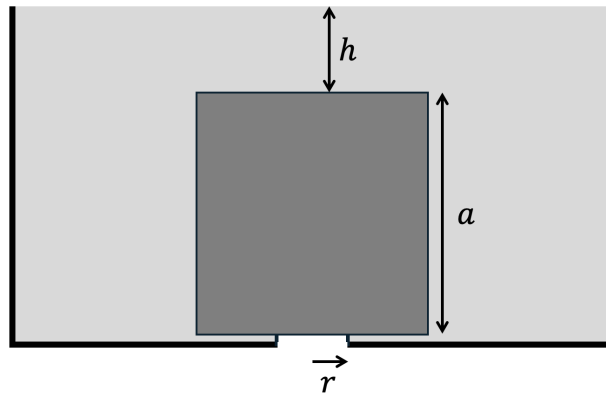


Figure 1: Wooden cube over a circular opening at the bottom of the vessel.

Deadline: May 28, 2026